

KOSZALIN, Poland

WMO: 121050

Lat: 54.2044N

Lon: 16.1553E

Elev: 114

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.3	(c) 16.1	(d) 4.2	(e) 6.9	(f) 12.7	(g) 9.7	(h) 9.1	(i) 17.9	(j) 25.0	(k) 42.5	(l) 22.5	(m) 41.8	(n) 5.0	(o) 150	(p) 0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.1	(c) 82.6	(d) 66.7	(e) 78.9	(f) 65.6	(g) 75.7	(h) 64.5	(i) 69.5	(j) 78.8	(k) 67.4	(l) 75.4	(m) 65.8	(n) 73.1	(o) 7.6	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 66.3	(b) 97.2	(c) 72.8	(d) 64.5	(e) 91.3	(f) 71.0	(g) 63.0	(h) 86.4	(i) 69.5	(j) 33.6	(k) 79.3	(l) 31.9	(m) 75.3	(n) 30.7	(o) 73.3	(p) 77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.0	(b) 17.1	(c) 14.8	(e) 5.7	(f) 89.6	(g) 7.6	(h) 3.5	(i) 0.2	(j) 92.2	(k) -4.3	(l) 94.2	(m) -8.6	(n) 96.2	(o) -14.1	(p) 98.7		
			DB													
			WB													
				5.7	89.6	7.6	3.5	0.2	92.2	-4.3	94.2	-8.6	96.2	-14.1	98.7	
				4.9	73.2	7.5	2.3	-0.5	74.8	-4.9	76.1	-9.1	77.4	-14.6	79.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	31.9	33.4	38.0	46.5	53.8	59.5	63.9	64.2	57.5	49.1	40.8	34.5
	DBStd	13.29	8.19	7.65	6.85	7.55	7.15	5.87	5.48	5.21	5.30	6.05	6.15	7.81
	HDD50	2450	563	466	376	155	37	1	0	0	4	91	278	479
	HDD65	6414	1028	886	838	556	357	186	88	79	233	494	725	944
	CDD50	1661	0	0	3	51	155	287	432	440	228	62	3	0
	CDD65	149	0	0	0	2	10	22	54	54	7	0	0	0
	CDH74	1088	0	0	0	21	99	199	408	327	34	0	0	0
	CDH80	245	0	0	0	1	15	54	107	65	3	0	0	0
Wind	WSAvg	7.0	8.1	7.9	7.7	7.1	6.5	6.8	6.3	6.0	6.2	6.7	7.3	8.1
Precipitation	PrecAvg	27.9	1.8	1.2	1.5	1.6	2.2	3.0	3.5	3.1	3.2	2.5	2.5	2.2
	PrecMax	39.8	3.9	3.0	3.4	4.4	4.4	9.3	8.3	5.8	8.7	9.1	6.3	6.1
	PrecMin	19.0	0.1	0.2	0.3	0.2	0.4	0.8	0.5	0.4	0.4	0.5	0.6	0.2
	PrecStd	4.9	0.9	0.7	0.8	1.0	1.0	1.9	1.9	1.5	1.8	1.8	1.4	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.7	52.2	61.2	77.6	82.1	86.4	88.2	86.8	78.1	68.2	56.7	52.5
		MCWB	47.2	47.1	49.7	59.7	64.2	68.8	69.6	69.0	65.1	59.1	53.3	50.4
	2%	DB	46.8	48.5	55.3	69.0	76.7	79.5	82.9	80.8	73.6	63.6	53.3	49.4
		MCWB	44.4	44.9	47.2	55.5	62.2	66.1	67.8	66.3	62.4	57.6	50.2	47.2
	5%	DB	44.2	45.9	51.0	63.6	71.2	74.6	78.4	77.4	69.8	60.6	51.2	46.6
		MCWB	42.4	43.0	45.7	52.2	59.1	63.4	66.5	65.6	61.1	56.1	48.7	44.5
	10%	DB	41.9	43.4	47.4	59.2	66.7	70.3	74.7	74.2	66.2	57.8	49.1	44.2
		MCWB	40.1	41.0	43.3	50.3	57.4	61.5	65.0	64.5	59.3	54.4	47.0	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	48.5	52.1	60.5	66.4	71.2	72.3	71.2	66.9	61.0	54.3	50.9
		MCDB	50.4	51.2	57.9	75.1	77.1	83.7	82.6	81.0	74.2	65.5	56.1	52.0
	2%	WB	44.8	45.8	48.7	56.5	63.5	67.0	69.8	68.8	64.1	58.5	50.9	47.5
		MCDB	46.5	48.1	53.5	67.2	73.6	76.4	79.7	76.8	70.5	62.3	52.8	49.1
	5%	WB	42.4	43.3	46.0	53.5	60.9	64.7	67.7	67.1	61.9	56.5	48.9	44.7
		MCDB	44.0	45.3	50.6	61.7	69.6	72.8	76.0	74.3	67.3	59.8	50.8	46.3
	10%	WB	40.3	41.0	43.6	51.0	58.2	62.4	65.9	65.5	60.2	54.7	47.0	42.6
		MCDB	41.8	43.4	47.0	57.9	65.5	69.0	72.6	72.3	65.2	57.6	48.9	44.1
Mean Daily Temperature Range	5% DB	MDBR	6.9	8.5	11.1	14.7	15.5	14.4	14.1	14.3	13.1	10.7	7.4	6.6
		MCDBR	7.7	11.1	17.2	22.1	22.5	21.8	21.2	19.9	18.6	14.3	9.5	8.0
		MCWBR	7.1	8.6	11.1	11.8	11.7	11.3	10.0	9.7	9.5	9.3	7.6	7.4
	5% WB	MCDBR	7.6	10.0	15.4	19.7	20.4	19.6	18.6	17.0	15.7	12.9	9.1	7.7
		MCWBR	7.1	8.3	10.6	11.3	11.5	11.3	10.0	9.6	9.3	9.0	7.7	7.4
Clear-Sky Solar Irradiance	taub		0.312	0.338	0.365	0.403	0.401	0.396	0.420	0.418	0.378	0.361	0.340	0.309
	taud		2.303	2.306	2.283	2.244	2.306	2.349	2.300	2.312	2.401	2.407	2.347	2.301
	Ebn at Noon		201	236	255	260	268	271	261	253	250	226	188	176
	Edh at Noon		18	26	32	38	38	37	38	36	29	23	18	15
All-Sky Solar Radiation	RadAvg		157	364	758	1320	1697	1796	1640	1369	986	503	198	111
	RadStd		21	53	81	159	190	134	192	125	106	75	28	11

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	N/A
(47) Regional (35 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-234	+106	N/A

Nomenclature: See separate page